

Linear A KU-PA and cyperus: a sign-group family and the word behind Greek κύπαιρος

Alberto Acedo Biome Makers Inc. Draft v0.4, 5 September 2026. Not for circulation.

Abstract

The sign-group KU-PA (AB 81-03) and the forms built on it (KA-KU-PA, KU-PA-ZU, KU-PA-JA, KU-PA-RI, and the ligature *304+PA-KU-PA) are examined on distributional grounds before any phonetic comparison is attempted. KU-PA recurs within single documents with distinct small integer quantities (ZA 11a, ZA 11b, KH 29); its derivatives stand next to the logogram *303, conventionally read as cyperus, on two roundels from Haghia Triada (HT Wc 3015, 3016) and on KH 5; the base word follows a commodity logogram on the nodule HT Wa 1020; and HT 16, headed by KA-KU-PA, has the format of the Mycenae Ge spice records. Two of the family's affixes, the prefix KA- and the suffix -JA, recur elsewhere in the corpus above the level expected under permutation, and their distribution over document types can be described. Only after this is the sign-group compared with Greek κύπαιρος / κύπειρον, a word of Pre-Greek origin according to Frisk and Beekes, and with Mycenaean ku-pa-ro, which glosses the cyperus logogram at Knossos (KN Ga 517) in the same way that KA-KU-PA accompanies *303 at Haghia Triada. Younger's reading of KA-KU-PA as the name of a contributor is discussed; the sealing groups of the Haghia Triada roundels and the structure of KH 5 favour a party associated with cyperus deliveries rather than the commodity itself. The proposal is therefore twofold: the stem KU-PA is the Minoan word from which the Greek name of the plant derives, and the family attested in the archives is either that word or, more probably, a name of a person, place or estate formed from it. The points at which either reading could fail are stated.

1. Introduction

The interpretation of Linear A sign-groups rests on a small number of secure or widely accepted identifications obtained from the internal analysis of the documents: KU-RO 'total' and KI-RO 'deficit' (Duhoux 2011) [1], a set of transaction terms and headings (SA-RA₂, KA-PA, A-DU; Schoep 2002 [2]; Montecchi 2010) [3], and the toponyms and anthroponyms that recur in Knossian Linear B (Godart 1984) [4]. Identifications proposed on phonetic grounds alone, whatever the language of comparison, have not gained acceptance, for reasons set out by Duhoux (1989) [5] and confirmed by recent computational work: with a syllabary of some ninety signs and sign-groups of two or three syllables, any sufficiently large lexicon yields matches (Nepal and Perono Cacciafoco 2024) [6].

That observation can be quantified. For the 729 Linear A sign-groups with at least two consonants, read with the values of their Linear B homographs and reduced to a consonantal skeleton on a common inventory of thirteen classes, the proportion of skeletons attested in a comparison lexicon is 47% for Mycenaean Greek, 43% for Hittite, 62% for Ugaritic, 70% for Hebrew and 50% for Etruscan; against each lexicon with its consonants permuted, the ratios are 1.47, 1.46, 1.50, 1.68 and 1.57 respectively. No comparison language stands out, and the highest

ratio corresponds to the largest lexicon. A method that starts from sound has no discriminating power on this corpus (Fig. 1).

Fig. 1. Proportion of Linear A consonantal skeletons attested in five comparison lexicons, against permuted lexicons.

The procedure followed here is therefore the one that produced KU-RO: the function of a sign-group is established from its behaviour in the documents, and external comparison is admitted afterwards, as corroboration. Two additions are made. First, each distributional claim is tested against a permutation null (fixed-margin randomisation for co-occurrence, permutation of the affixed syllable for morphology); the same tests were applied to Etruscan, where the genitive before clan and the northern distribution of the marked sibilant are recovered (58% against a base rate of 26%; 0.6 against 0.1). Second, attention is paid to a behaviour that in an inventory separates a counted item from a name: recurrence within a single document with different quantities. The purpose of the calibration is the ordinary one: an instrument is tried first on material whose answer is known. Applied to Etruscan, where the genitive in -s or -l before clan and the northern distribution of the marked sibilant are established facts, the same tests recovered the genitive (58% of the words before clan, against a base rate of 26%), recovered the loss of internal vowels as a change over time that survives control by city, and dismissed an apparent change over time in the sibilant once city was controlled. Each null is a permutation of the data that preserves what must be preserved (the number of sign-groups in each document and the number of documents in which each sign-group occurs; or the inventory of final syllables), so that a claim of association is measured against what the same material would produce by chance. The procedure, its calibration and its negative results are described in a separate paper; the present note reports a single result.

2. The evidence

2.1 KU-PA

The readings below follow GORILA as tabulated in the LinearA Explorer (Hogan 2022) [7]; every attestation of the sign-group and of its derivatives has been checked against the SigLA dataset (Salgarella and Castellan 2020) [8], which confirms AB 81 followed by AB 03 in all of them. The eleven documents in which KU is followed by AB 56 (pa₃) belong to a different sign-group, KU-PA₃-NU, a personal name with a Mycenaean counterpart (ku-pa-nu-we-to), and to KU-PA₃-RI-JA (HT 24a), on which see 2.3.

Document	Site	Support	Context
ZA 11a	Zakros	tablet	QE+GRA+PA 4 KU-PA · GRA+PA 1 PI-*310-A OLIV 2
ZA 11b	Zakros	tablet	I-OLIV 1 I-DE RA

Document	Site	Support	Context
			2 KU-PA · GRA+PA 3
KH 29	Khania	tablet	RA · TA ₂ 1 KU- PA 2 *303 ½
HT 110a	Haghia Triada	tablet	SI-DU-*34-KU-MI *303+E 20 KU-PA 1 KU-RO 100

KU-PA is written on both faces of ZA 11 with the grain sign GRA+PA and the quantities 1 and 3, and at Khania with the quantity 2 in a document that also records *303. The quantity is in every case a small integer; no fraction follows the sign-group, although fractions follow the commodities recorded around it. Recurrence within a document is not in itself diagnostic of a commodity: the sign-groups that recur within documents are transaction terms (KU-RO, KI-RO), headings (MA-RU-ME on HT 24a, with 10 and 6) and, occasionally, names (DA-KU-SE-NE on HT 103, with 6 ½ and 1; KU-PA₃-NU twice on HT 122a). What the recurrence shows is that KU-PA is entered with quantities in more than one place in the same document; this is compatible with a counted commodity and with a party to whom grain is allotted twice. It must also be noted that the bare sign-group precedes a grain sign on ZA 11 and its own quantity on KH 29 and HT 110a; the association with *303 is carried by the derivatives, not by KU-PA itself, which may qualify a grain entry or record a particular presentation of the product (Figs. 3, 4).

Fig. 3. The attestations of KU-PA on ZA 11a, ZA 11b, KH 29 and HT 110a: sign sequence after SigLA in normalised forms, with the GORILA transliteration. Author's schematic, not a facsimile.

*Fig. 4. The signs *303, *626, *304, and the sign-groups KU-PA, KA-KU-PA and KU-PA₃ in standard form.*

2.2 The derivatives and the logogram *303

Sign-group	Document	Context
KA-KU-PA	HT Wc 3015 (roundel)	KA-KU-PA *303
KA-KU-PA	HT Wc 3016 (roundel)	KA-KU-PA *304+PA- *303+D
KA-KU-PA	HT 16 (tablet, heading)	KA-KU-PA · DI-NA-U ⅛ WA ⅓ *188+KU ⅔ SA-PO

Sign-group	Document	Context
		ZE $\frac{1}{16}$
KU-PA-ZU	KH 5 (tablet)	KU-PA-ZU *303 3 $\approx \frac{1}{6}$ NI 2
*629 KU-PA	HT Wa 1020 (nodule)	*629 KU-PA $\frac{1}{4}$ JE-TA-NA
KU-PA-JA	HT 116a (tablet)	U-TA-RO · TE · KU-PA-JA GRA 16 PU-RA ₂ · GRA 40
KU-PA-RI	PE 1 (tablet)	U-KA-RE-A-SE-SI-NA KU-PA-RI VIR 50 GRA+PA 26 $\frac{1}{2}$

Two roundels from Haghia Triada bear KA-KU-PA followed only by *303 or by its ligature; KU-PA-ZU precedes *303 with a quantity at Khania; On the nodule HT Wa 1020 the sign-group follows the commodity logogram *629 (*304+PA) as two separate elements (SigLA reads *629, KU, PA): the nodule pairs a commodity sign with the sign-group rather than ligaturing them, and the transliteration *304+PA-KU-PA of the LinearA Explorer should not be read as a ligature. Of the twelve sign-groups that precede *303 anywhere in the corpus, two belong to this family. None of the family's attestations involves wine, oil, livestock or a vessel sign (Fig. 2).

Fig. 2. Documents, logograms and quantities associated with KU-PA and its derivatives.

The reading of *303 (A 303) and of its ligatures (A 626, A 629) as cyperus is the usual one (Palmer 1995 [9], apud Younger) but is not secure: Del Frio, Nosch and Rougemont (2010) [10] gloss A 626 as “cyperus / grain (?)”, and Schoep (2002, 93 [2], apud Younger) records the disagreement between Hallager and others on which sign represents the plant. The argument below uses the conventional reading and, if accepted, supports it; the circularity is acknowledged and is broken only by the external comparison in section 3.

HT 16 deserves separate mention. Its entries are single signs used as commodity abbreviations followed by small fractions (WA $\frac{1}{3}$, *188+KU $\frac{3}{6}$, SA-PO ZE $\frac{1}{16}$), a format otherwise found in the Linear A tablets only on HT 20 (WA $\frac{1}{4}$, *188+KU $\frac{1}{2}$) and, with logograms, on HT 23a, HT 32, HT 35 and HT 60. The format is that of the Mycenae Ge series, in which condiments are recorded by acrophonic abbreviation with the smallest measures of the system. A related observation concerns oil. The oil logograms of Linear A divide into two groups by their quantities and their company. OLE+U, plain OLE, OLE+KI, OLE+MI, OLE+DI and OLE+TU are recorded in whole units (medians between 3 and 7, up to 54) and share their documents with grain, olives, figs and *304, above chance in every case (hypergeometric tests on 1,719 documents): they are the oils of the estate's produce. OLE+TA, OLE+RI and OLE+NE are recorded in fractions and share their documents with *303 (OLE+TA in 4 of 7 documents, $p < 0.001$; OLE+RI and OLE+NE in 2 of 4 each, $p = 0.006$), with *308, MI+JA+RU and the

ligatures QA₂+[*?]+PU and QA₂+[*?]+RE, and hardly ever with grain or olives: HT 23a, HT 30, HT 32, HT 35, HT 60, KH 61 and KH 85 belong to this group, and on HT 30 and HT 100 the allocation heading SA-RA₂ is followed by *303 with OLE+TA. The second group corresponds in function to the qualified oils of the Pylos Fr series (ku-pa-ro-we, wo-do-we, pa-ko-we), with the qualifier written as a ligatured sign rather than as a word, a device also used by Linear B in CYP+KU and PYC+KU; the values of TA, RI and NE are not known, and OLE+TA, the one most often recorded with *303, is the natural candidate for cyperus-scented oil. The documents in which the KU-PA family occurs are thus records of an administration of aromatics and scented oils, and the document headed by KA-KU-PA is, by its format, a list of condiments.

2.3 Younger's reading

Younger's commentary on HT 16 already connects KA-KU-PA with *303, but in the opposite direction: KA-KU-PA, which "appears on two roundels, HT Wc 3015 and 3016, contributing *303", is taken to be "the person responsible for the contributions of *303". The Linear A Lexicon (Younger 2024a) [11] glosses KU-PA as "name in a list?, transaction term?" (HT 110a.2, HT We 1020a, ZA 11a.5, b.3), KU-PA[on KH 29 as "name in a list", KA-KU-PA as "(person's name?)", KU-PA-JA as "name in a list", KU-PA-ZU as "third name in a three-name list", KU-PA-RI as "word (placename?) in a list", and KU-PA₃-RI-JA as "(from KU-PA-RI) heading, first word in a three-word list". Neither cyperus nor ku-pa-ro is mentioned. Younger's own hesitation over the base word, between a name and a transaction term, is resolved here towards the second alternative; and his derivation of KU-PA₃-RI-JA from KU-PA-RI implies an alternation of pa and pa₃ within the family, so that the sign-level distinction between KU-PA and KU-PA₃-NU holds for the signs while KU-PA₃-RI-JA, a heading, may belong with KU-PA-RI.

Whether the sign-group on a roundel names the commodity or the party to the sealed transaction is a question discussed by Hallager (1996) [12], on whose analysis Younger's reading depends. A roundel reading KA-KU-PA *303 is as naturally read as commodity as it is as contributor. The seal impressions do not decide the matter either: at Haghia Triada one seal, CMS II,6 no. 142, stamps the roundels bearing PA-SE-JA (HT Wc 3001, 3002), KA-RE-RO, ZU-NI-KA, TA-NI-KA, WA-O-MI, WA-DI-NI and A333-DI-NA-SU-KA alike (data from SigLA), so that the seal identifies the receiving official, as Hallager showed, and not the party or commodity named. The prefix KA- does not decide the matter (2.4); the behaviour of the base word does: it is counted, it stands beside a commodity sign on a nodule, its derivative KU-PA-ZU is followed by *303 and a quantity, and it never appears in a personnel list or with VIR. Two further observations bear on the choice. First, the roundels of Haghia Triada fall into two sealing groups (Weingarten 1987 [13]; Hallager 1996 [12]; data from Younger's commentary and SigLA): those stamped by CMS II,6 no. 142 carry sign-groups without commodity logograms (PA-SE-JA, KA-RE-RO, TA-NI-KA, WA-O-MI, WA-DI-NI, ZU-DI-RA), while those stamped by nos. 11, 21, 77 and 148 carry a sign-group with a commodity logogram: KA-KU-PA with *303 (no. 11, three impressions) and with *629 and *624 (no. 21, two impressions), QA-KU-RE with *303 ligatures (no. 21, five impressions), A-RA-TU-ME with OVIS f (no. 21, six impressions), KU-MI-NA-QE with CAP (no. 77). In that group the sign-group is naturally the party and the logogram the commodity,

since A-RA-TU-ME with ewes or KU-MI-NA-QE with goats can hardly be commodity words. Second, the entries of KH 5 are built as sign-group, logogram, quantity: WI-SA-SA-NE *626 2, KU-PA-ZU *303 3; two different sign-groups before two cyperus logograms in one tablet are more easily two parties than two words for the plant. These patterns favour Younger's reading of the family's function. They do not affect the comparison in section 3: a party named after the plant, a person, a place or an estate producing it, is what a name formed on the stem KU-PA would be, and Greek itself names places after plants (Kyparissos, Marathon, Selinous). The two readings are then compatible: KA-KU-PA a party named after the plant, KU-PA the plant's name. The pattern is not isolated. KU-MI-NA-QE, which heads HT 89 and stands with CAP on the roundel HT Wc 3014 in the same sealing group as KA-KU-PA, shares its first two syllables with Mycenaean ku-mi-no, cumin; and KI-KI-RA-JA, which heads the personnel list of HT 85b, is formed with the suffix -JA on KI-KI-NA, the sign-group that Neumann (1960) [14] identified with the sycamore fig and that is entered after the fig logogram on HT 88 (NI · KI-KI-NA 7). A screen of the whole vocabulary against twenty-six plant names of Pre-Greek or loan origin known from Mycenaean yields 37 stem matches against 18.6 expected when syllables are permuted (maximum 32 in 200 permutations); most of the excess is the KU-PA family itself, and two-syllable matches are individually weak, so the parallels are offered as a pattern to be tested, not as readings.

2.4 The affixes

Two affixes of the family recur in the corpus above the level expected under permutation of the affixed syllable. The prefix KA- forms five pairs of attested stem and prefixed form against 1.7 expected ($p = 0.013$): KA-KU-PA / KU-PA, KA-SA-RU / SA-RU, KA-NU-TI / NU-TI, KA-I-KA / I-KA, KA-RE-RO / RE-RO. The prefixed forms occur on roundels (HT Wc 3015, 3016, 3003; KN Wc 3) or in a tablet heading (HT 16); the bare forms are list entries with quantities (SA-RU 20 on HT 86, I-KA 5, NU-TI with VIR). The preference for roundels holds for the corpus as a whole: sign-groups beginning with ka- make up 17% of roundel sign-groups (5 of 29) against 4% of tablet sign-groups (29 of 697). The bases are names or place-names as often as commodities (KA-SA-RU is the form that survives in Knossian Linear B as ka-sa-ro), so KA-marks the document type rather than the class of referent; Younger (2010) [15] had already noted prefixation and suffixation as the characteristic feature of roundel sign-groups. The suffix -JA forms seven pairs against 3.1 expected ($p = 0.020$): KU-PA-JA / KU-PA, PA-SE-JA / PA-SE, A-MA-JA / A-MA, A-RI-JA / A-RI, A-SE-JA / A-SE, JA-SA-JA / JA-SA, PU₂-RE-JA / PU₂-RE; in these pairs the bare form is a heading (PA-SE GRA+QE 20 on HT 18; A-MA on MA 1b and ZA 7b; A-SE on HT 81) and the suffixed form an entry with a quantity (KU-PA-JA GRA 16; PA-SE-JA 20 with VIR; A-MA-JA *303 6). The 49 sign-groups ending in -ja taken as a whole show no such distribution (41% in headings against a base rate of 34%), so the observation is confined to the pairs. -ZU, -RI and -RI-JA are too rare to test. The family may thus be described as a stem KU-PA, a roundel or heading form KA-KU-PA, a derived form KU-PA-JA used as an entry, and the longer forms KU-PA-RI and KU-PA₃-RI-JA.

3. Comparison

Mycenaean cyperus has its own logograms, *124 PYC and *125 CYP, and the sign-group ku-pa-ro accompanies them in the records of aromatics. In the corpus used here: KN Ga(1) 517 tu-wi-no / ku-pa-ro AROM+PYC 1; KN Ga(1) 519 ku-pa-ro *171 2; KN Ga(3) 465 ni-so-de / ku-pa-ro T 1; PY Un 267 ku-pa-ro₂ AROM 6 (the perfumer's allocation); PY Un 249 ku-pa-ro₂ AROM 2 T 5; PY Fr 1203 ku-pa-ro-we wo-do-we OLE (see DMic s.v. ku-pa-ro [16]; Aurora 2015) [17]. KN Ga 517, in which the word is followed by the cyperus logogram and a quantity, is the counterpart of HT Wc 3015 and 3016. The Linear B logogram itself carries the syllable KU as an acrophonic element in CYP+KU and PYC+KU; Linear A has, on the nodule HT Wa 1020, the commodity sign *629 followed by KU-PA (*304 being a grain sign, not *303). If *303 is the antecedent of *125, as the conventional reading assumes, the pairing of word and logogram is inherited rather than analogical.

Greek κύπειρον, κύπειρος, Ionic κύπερος, and κύπαιρος in Alcman denote the galingale, *Cyperus longus* and *C. rotundus*; Hesychius glosses κύπερα as ropes plaited from it. Frisk (GEW s.v.) [18] regards the word as a foreign one of unknown origin whose fluctuating form reflects borrowing, and refers to Mycenaean ku-pa-ro; Beekes (2010 s.v.) [19] classes it as Pre-Greek. The correspondence with the Linear A family is as close as the syllabary allows: KU-PA is the stem, KU-PA-RI (PE 1) supplies the -r- of κυπαρι-/κυπερ-, and KU-PA₃-RI-JA, if Younger's derivation is accepted, supplies the form with yod reflected in ku-pa-ro₂ (kuparyos). The vowel fluctuation of the Greek forms is what would be expected from a source with an unstable second vowel. The identification of a Minoan plant name through a Greek gloss has precedents in Neumann (1960, 1962) [14]; the Minoan loan stratum in Mycenaean Greek was surveyed by Renfrew (1998) [20]; and the order of argument, function first and etymology afterwards, is that of Piquero (2015) [21] for pa-ra-ku-ja, where a Minoan channel is also assumed. What is added here is not the etymology, which the dictionaries already give, but the Minoan sign-group itself, found in the Minoan documents beside the corresponding logogram, with a morphology that accounts for the Greek variants; whether the archives use the stem as the name of the plant or as the name of a party associated with it, the stem is the same.

An alternative comparison must be considered. κύπελλον 'cup', with κύπη, κύμβη, Latin cūpa and Sanskrit kūpa, is a widely diffused word for containers, also regarded as Pre-Greek, and would suit KU-PA as a counted vessel or unit. Nothing in the context of KU-PA or of its derivatives is a vessel: the vessel names of Linear A are those of HT 31, written beside their pictograms, and none is built on ku-pa. The distribution favours the plant.

4. Discussion

The proposal has three supports: the consistent association of the family with the logogram *303 and with the documents of an aromatics administration, a morphology that can be described from the corpus, and an external form with the appropriate history. On the function of the family the evidence is divided: the recurrence of KU-PA with quantities and the format of HT 16 are compatible with a commodity, while the sealing groups of the roundels and the structure of KH 5

favour a party associated with cyperus deliveries, as Younger proposed. The reconciliation offered here is that the family is a name formed on the plant's word. It is not a phonetic argument and has no bearing on the affiliation of the Minoan language. Its weakest points are the reading of *303, which is conventional rather than secure, and the fact that the bare sign-group is followed by a grain sign on ZA 11. The proposal can be tested: KU-PA and its derivatives should continue to appear with *303 and with dry commodities, in integer quantities, and not as a container of liquids; a new roundel with KA-KU-PA should carry *303 or a ligature of it; KU-PA-RI, if attested again, should accompany cyperus. Sesame (SA-SA-ME, HT 23a, recorded with the fraction $\frac{1}{16}$ in a document that also lists *303 and qualified oils; cf. Mycenaean sa-sa-ma in the Ge series) is the only other commodity term in the corpus that meets the same conditions; it is treated separately.

Data

Transliterations: GORILA (Godart and Olivier 1976-1985) [22] as tabulated by G. Douros and published in machine-readable form by the LinearA Explorer (Hogan 2022) [7]. Sign-level verification: the SigLA dataset (Salgarella and Castellan 2020 [8], CC BY-NC-SA 4.0), in the form redistributed by pyaegaeon (sigla-corpus-v1). The Explorer labels the Zakros tablets, including ZA 11, as stone vessels; the support has been corrected from SigLA. Statistical tests and scripts accompany the note.

Declaration of assistance

Corpus processing, permutation tests and a first draft of this text were produced with the assistance of a language model (Claude, Anthropic) under the author's direction. All readings, comparisons and claims have been checked by the author, who is responsible for them.

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